Order and entropy of lattices of unit vectors in d dimensions: label-free discovery of unusual local organization in simulated lattices and synthetic images

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Abstract

Averaged order parameters say how ordered a lattice of orientational states is, not where its organization is unusual. We extend the local order and entropy quantities of the MyUncle maintenance-lattice model to unit vectors in d dimensions (von Mises–Fisher distribution); they equal MyUncle's at $d = 2$. On this basis we build a label-free score that judges how rare the state of a site is at several prevalence scales and smooths it over the lattice, and compare it with four baselines, with settings frozen before the held-out data. Smoothing the score over the lattice raised spatial coherence on synthetic images and, with DINOv2 embeddings, average precision, but it lowered average precision on simulated MyUncle lattices; in a controlled study on an independent generator it hurt at every anomaly size, least for compact anomalies. A supervised search over 360 configurations with GaaR, an evolutionary search framework, raised the lower-tail average precision on fresh test scenarios from 0.18 to 0.26, but the paired interval of the difference included zero and random search also beat the default, so any gain belongs to the selected configuration, not to the search. All evidence is simulated; code and results are open.

Keywords: lattice models, von Mises–Fisher distribution, orientational order, entropy, unsupervised anomaly detection, spatial smoothing

1. Introduction

Many pattern recognition problems put a vector on every site of a grid: orientation fields of ordered media, and embeddings of the patches of an image. Averaged order parameters say how ordered such a lattice is, not where its organization is unusual. Finding unusual local organization without labels, and saying how unusual it is in the language of order and entropy, is the problem addressed here. Unsupervised anomaly detection has been developed in other fields: in astronomy for large surveys (Baron and Poznanski, 2017; Lochner and Bassett, 2021), and in medical imaging mostly as detection of deviation from a healthy appearance with generative models (Schlegl et al., 2019).

MyUncle (Sandler, 2026a) is a simulation framework for a maintenance lattice: rotors on a square lattice whose order is eroded by a state-dependent flux and restored by repair, with an order-entropy bridge that links the length of the neighbors' mean unit vector to a von Mises entropy. This paper makes four contributions. (1) It extends the local order and entropy quantities from the plane to unit vectors in d dimensions (the von Mises–Fisher distribution) and verifies that they equal MyUncle's at $d = 2$ against the unmodified code. (2) It builds a label-free score that judges the rarity of the state of a site at several prevalence scales and smooths it over the lattice. (3) It tests the score, with frozen settings and four baselines, on lattices simulated with MyUncle (damaged sites as ground truth) and on synthetic stained-tissue images, where the sites are image patches described by an embedding; the results are mixed and are reported as such. (4) It studies a supervised configuration search with GaaR (Sandler, 2026c) on fresh test scenarios, and provides an open pipeline (Sandler, 2026b). All evidence is simulated; no experimental image was analyzed.

2. Order and entropy of a lattice of unit vectors

Let u_j be the unit vector at site j of a lattice, N_i the valid neighbors of site i (four, on a square lattice), and d the dimension of the vectors. MyUncle's local disorder is one minus the length of the mean of the neighbors' unit vectors; the same definition holds in any dimension:

$$R_i = \left| \tfrac{1}{|N_i|} \textstyle\sum_{j \in N_i} u_j \right|, \qquad D_i = 1 - R_i. \tag{1}$$

The concentration κ_i of the von Mises–Fisher distribution on S^{d-1} (Banerjee et al., 2005; Mardia and Jupp, 2000) with mean resultant length R_i solves

$$A_d(\kappa) = \frac{I_{d/2}(\kappa)}{I_{d/2-1}(\kappa)} = R_i, \tag{2}$$

and the entropy of that distribution and the information held in the local order are

$$\begin{aligned} S_{\mathrm{vMF}}(\kappa, d) = &-(\tfrac{d}{2} - 1)\ln\kappa + \tfrac{d}{2}\ln(2\pi) \\ &+ \ln I_{d/2-1}(\kappa) - \kappa A_d(\kappa), \end{aligned} \qquad H_\chi = S_{\mathrm{unif}} - S_{\mathrm{vMF}}, \tag{3}$$

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with S_{unif} the entropy of the uniform distribution on the sphere. For $d = 2$ these are MyUncle's $U = A(\kappa) = I_1/I_0$, $S(\kappa)$ and H_χ. The tests compare them with the vendored, unmodified core (Fig. 1): the local disorder on a periodic lattice agrees to 10^{-12} and the entropy to 10^{-9}. With a neighborhood of four sites the resultant length is a noisy estimate of the local order; this is a stated limitation.

3. A label-free score on the lattice

Embeddings. For images, a patch is a 224×224 px window (0.5 μm/px) of the image; it is encoded either by DINOv2 ViT-S/14 (Oquab et al., 2023) (384-d, pretrained on natural images, no fine-tuning) or by a deterministic handcrafted encoder of color, gradient-orientation and texture statistics (64-d) built on color deconvolution (Ruifrok and Johnston, 2001). Embeddings are centered, reduced by a cohort PCA (16 components, scikit-learn; Pedregosa et al., 2011) and normalized to unit length. For simulated lattices the vector of a site is the rotor $(\cos\theta, \sin\theta)$, optionally with the local disorder D as a third coordinate; the intact flag is never part of it.

Score. Appearance rarity p_{emb} is the cohort percentile of the mean cosine distance to the k nearest sites of a reference sample of the cohort (same-image neighbors within six lattice steps excluded). Because the prevalence of an interesting state is not known in advance, it is evaluated with k equal to 0.5, 1, 2, 5 and 10% of the reference sample and the highest rank counts. Two further rank-based axes, state-wise disorder p_{het} (the rank of D_i among sites of the same k-means prototype state) and contact rarity p_{adj} (the rarity of the states a site touches), are diagnostics. The novelty of a site is the mean over its 3×3 lattice window W_i of a weighted probabilistic OR of the axes,

$$N_i = \tfrac{1}{|W_i|} \textstyle\sum_{j \in W_i} q_j, \quad q_j = 1 - \prod_a (1 - p_{a,j})^{w_a}, \qquad (4)$$

with the frozen weights $(w_{\text{emb}}, w_{\text{het}}, w_{\text{adj}}) = (1, 0, 0)$: the default score is the smoothed multi-scale appearance rarity. The disorder and contact axes were part of the original design; they are kept as options because their effect depended on the embedding (Section 5). Candidates are the top fraction of N_i; connected candidates form regions, clustered across images into ranked findings with explanations made of measured numbers. Sandler (2026b) gives the pipeline, including a bridge that simulates any MyUncle preset and analyzes its states.

4. Evaluation design

All settings of the score were frozen on development data (`results/frozen_settings.json`, written before the held-out data existed), and the held-out data were evaluated once. The baselines are isolation forest, local outlier factor, nearest-neighbor distance ($k = 10$) and Mahalanobis distance on the same embeddings, each raw and smoothed in the same way as the proposed score (percentile rank, then the 3×3 window); implementations are from scikit-learn. Metrics are AUROC and average precision (AP) against the ground truth; for the top 3% of scores on images, the spatial coherence (the share in connected groups of at least three patches) and the region recall (planted instances containing a candidate).

Table 1: Simulated MyUncle lattices, damaged sites as ground truth. Held-out presets: AUROC and AP with the rotor embedding (columns 2, 3) and the rotor + D embedding (4, 5); AP on the development preset (6, 7).

Method	AUROC	AP	AUROC	AP	AP_{dev}	AP_{dev}
MyUncleVision (frozen, smoothed)	0.65	0.36	0.77	0.49	0.13	0.28
Isolation forest	0.72	0.59	0.75	0.50	0.45	0.36
+ smoothing	0.65	0.36	0.76	0.49	0.13	0.27
Mahalanobis distance	0.76	0.63	0.76	0.44	0.48	0.23
+ smoothing	0.71	0.40	0.75	0.47	0.16	0.26
Nearest-neighbor distance	0.72	0.56	0.72	0.38	0.42	0.20
+ smoothing	0.65	0.36	0.75	0.47	0.13	0.25
Local disorder D alone	0.63	0.32	0.58	0.26	0.12	0.10

Simulated lattices. MyUncle's kinetic Monte Carlo (unmodified core) was run from a damaged start; snapshots at steps 20, 40 and 60 of 48×48 sites were analyzed as images whose sites are patches. The damaged sites (sites that lost their coupling) are the ground truth. The development set is one preset (damaged share 10%; four runs); the held-out set is three other presets (damaged share 30%, 26% and 8%; four runs each). The presets are model parameter sets that MyUncle labels with application names (here polymer photodegradation, liquid crystal and magnetic storage decay); they are not fitted to measurements.

Synthetic images. No real image was available. Synthetic stained-tissue images (4480×4480 px) with tubule-like rings, glomerulus-like discs, vessels, stain variation, planted anomalies and artifacts imitate what the method measures. A development set (8 images, 3.1% of patches anomalous) and a held-out set (16 images, three anomaly types never used in development, 0.8% anomalous, stain shifts) were written by the author of the method, which limits what they can show.

5. Results

5.1. Simulated lattices

Figure 2 shows one simulated run. With the rotor alone, the frozen score was no better than smoothed isolation forest or nearest-neighbor distance (held-out AP 0.36, AUROC 0.65) and clearly below the unsmoothed baselines (Mahalanobis AP 0.63, isolation forest 0.59; Table 1). Smoothing lowered the AP of every baseline on both sets (isolation forest 0.59 to 0.36 on the held-out set): the damaged sites are scattered, not compact regions that a 3×3 window preserves. Adding the local disorder as a feature raised the AP of the frozen score on both sets (development 0.13 to 0.28, held-out 0.36 to 0.49); on the held-out presets it was then within 0.02 of the best AP, tied with smoothed isolation forest, while on the development preset an unsmoothed isolation forest was clearly better (0.36 against 0.28). The disorder map alone ranked the damaged sites poorly (held-out AP 0.26), so the gain comes from combining it with the state of the site. Intervals (95%, resampling whole runs) are in the result files; they overlap for methods whose averages are

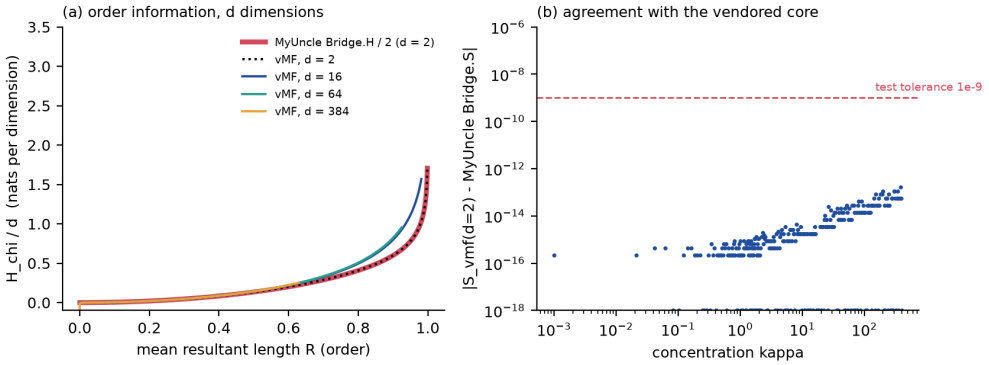


Figure 1: (a) The order-information bridge in d dimensions, per dimension: MyUncle's H_χ ($d = 2$, red) and the von Mises–Fisher generalization for $d = 2$, 16, 64 and 384. (b) Difference between the $d = 2$ entropy and the vendored MyUncle function.

close. The damaged share of three of the four parameter sets (about 10, 30 and 26%; the fourth is 8%) is near or above the largest prevalence scale of the rarity score (10%), a limit of the frozen settings that this testbed exposes.

5.2. Synthetic images

An earlier default (three-axis OR at one rarity scale, no smoothing) had an AUROC of 0.90 with handcrafted features and 0.22 with DINOv2, below chance: the planted states covered 3.1% of the patches, more than the 1% scale, and the DINOv2 embeddings of the planted regions are more homogeneous than normal ones, so the disorder and contact axes point the wrong way. Judging rarity at several scales fixed the appearance axis. On the development images, smoothing raised the spatial coherence of the frozen score from 0.42 to 0.91 (handcrafted) and from 0.30 to 0.87 (DINOv2), and smoothing raised the coherence of every baseline in all four cases; but a smoothed Mahalanobis baseline scored higher in AP than the frozen score (0.62 and 0.90 against 0.47 and 0.74, handcrafted and DINOv2).

On the held-out images with DINOv2 the frozen score was the best of all methods in AP (0.59, AUROC 0.99, coherence 0.65, region recall 1.00), ahead of smoothed Mahalanobis (0.35) and of the unsmoothed multi-scale appearance (0.15; Table 2). With handcrafted features it did badly (AP 0.04, region recall 0.29) although its AUROC was 0.80 and its coherence 0.89: the anomalies are small (about five patches), and a 3×3 window mixes each with normal neighbors and dilutes the ranks. The choice to smooth, made on the development images, therefore did not generalize across embeddings. The lattice axes behaved as on the development images: with smoothing they raised the AP with handcrafted features (0.26 against 0.04) and lowered it with DINOv2 (0.15 against 0.59), so they are diagnostics, not part of the default score. A coarser patch size was tolerated with DINOv2 and not with handcrafted features, and changing the smoothing or the axes changed the flagged set almost completely: these are design choices, not noise.

5.3. Configuration search with GaaR

The frozen configuration is one choice for all data, and the results above show it is not best for every embedding. GaaR (Goal-Aware Adaptive Regulation; Sandler, 2026c), an evolutionary search framework by the same author, was used to select a configuration of the score from 360 candidates: smoothing type and window (none, mean 3, 5, 7, Gaussian 5, 7, median 3, 5), five sets of axis weights, the PCA dimension (8, 16, 32) and the largest rarity scale (2, 5, 10%). This is supervised hyperparameter selection on labeled development scenarios, not label-free adaptation: the chosen configuration is fixed afterwards. The fitness is a lower-tail average precision, $F = \frac{1}{2}\overline{\mathrm{AP}} + \frac{1}{2}\min \mathrm{AP}$, over four development scenarios (the development images with each embedding and the development lattices with each of their two embeddings). All 360 configurations were evaluated once on these scenarios, so any search can be compared exactly with the optimum; each strategy of GaaR and plain random search then ran 100 times with a budget of 40 evaluations. The champions were evaluated once on fresh test scenarios that no search saw: 16 new synthetic images (new seeds, the held-out anomaly types) and 48 new lattice states, including a preset not used before (superconducting scaffold).

The default configuration ranked 64th of 360 on the development fitness (0.26; optimum 0.35). With 40 evaluations, the greedy and restart strategies found a configuration in the top 1% in 59% and 60% of the runs and random search in 25% (population: 46%), whereas at the top 5% level random search did as well or better (87% against 71%, 72% and 85%). On the test scenarios (Table 3) the champion of the restart and population strategies raised the test fitness from 0.18 (default) to 0.26, mainly by lifting the worst scenario (handcrafted-feature images, AP 0.04 to 0.17), while the AP with DINOv2 fell from 0.50 to 0.41. The paired 95% interval of this gain (+0.080) is [−0.05, +0.17] and the champion was better than the default in 89% of the resamples, so with 16 independent units per scenario it cannot be distinguished from zero. The champion of the greedy strategy and of random search (the same configuration here) reached 0.21, and the exhaustive development optimum 0.25: every development-optimized configuration beat the default, but their order did not follow the development fitness. The gain is therefore due to the selected configuration (a smaller PCA dimension and the disorder axis at weight 0.5, with 3×3 mean smoothing) and not to the search algorithm; each strategy

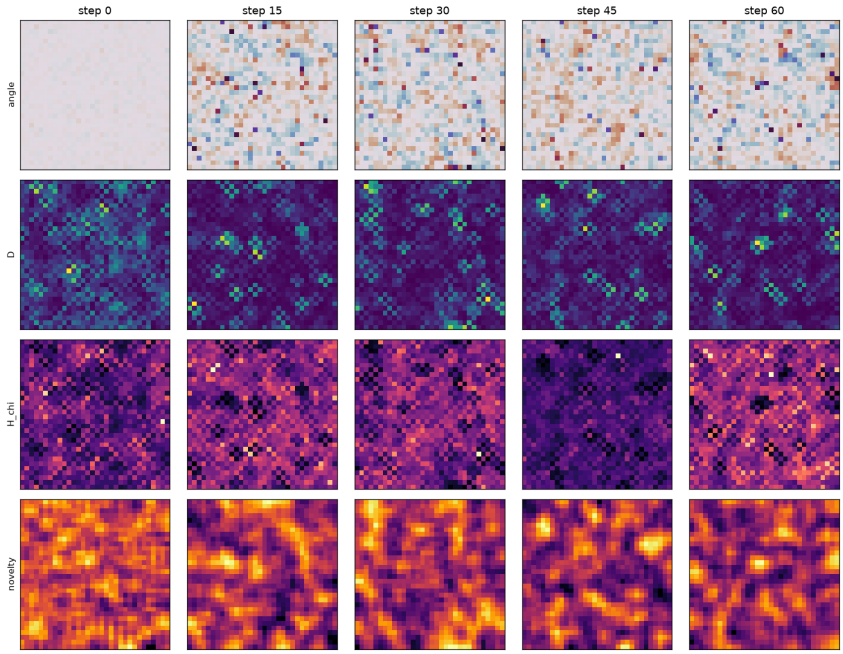


Figure 2: A MyUncle simulation analyzed with MyUncleVision (chiral scaffold preset, 32×32 sites). Rows: rotor angle, local disorder D, order information H_χ ($d = 2$) and novelty of the frozen score; columns: steps 0, 15, 30, 45 and 60. Novelty is a percentile rank relative to the snapshots shown and has not been validated against known defect locations.

Table 2: Held-out synthetic images (settings frozen beforehand, evaluated once): AUROC, AP, spatial coherence (coh.) and region recall (rec.), handcrafted features (left) and DINOv2 (right).

Method	AUROC	AP	coh.	rec.	AUROC	AP	coh.	rec.
MyUncleVision (frozen, smoothed)	0.80	0.04	0.89	0.29	0.99	0.59	0.65	1.00
Appearance, multi-scale, unsmoothed	0.98	0.17	0.31	0.86	0.95	0.15	0.20	0.86
Three-axis OR, smoothed	0.97	0.26	0.84	0.86	0.93	0.15	0.66	0.71
Isolation forest, smoothed	0.93	0.12	0.58	0.71	0.98	0.25	0.55	1.00
Mahalanobis distance	0.96	0.13	0.21	0.71	0.93	0.17	0.25	0.71
+ smoothing	0.93	0.11	0.59	0.86	0.99	0.35	0.58	1.00
Nearest-neighbor distance	0.96	0.14	0.28	0.57	0.43	0.01	0.03	0.14
+ smoothing	0.91	0.12	0.64	0.71	0.62	0.04	0.42	0.29

contributed one champion, four scenarios are few, and the test data come from the same generators as the development data.

A label-free alternative was explored. Moran's I of the unsmoothed rarity score (spatial autocorrelation of the score field) was below 0.02 in exactly the two lattice scenarios, one development and one test, in which smoothing lost most of the AP, and above 0.1 in the other six. A rule that smooths only above 0.05 would have avoided the largest loss in both, but it would not have helped the handcrafted-feature test images (eight scenarios, one threshold, exploratory).

5.4. Smoothing against anomaly morphology

The results above leave open when smoothing helps. A controlled study on an independent generator varied only the spatial size of the anomalies. Each of 50 lattices (32×32 sites, 16-dimensional embeddings) has smooth background domains and 3% anomalous sites with a fixed separability (a shift of 10 noise standard deviations, calibrated beforehand on the unsmoothed score only); the anomalies are $b \times b$ blocks, $b = 1$ being scattered single sites and $b = 6$ compact regions, and each block has its own direction, so no recurring state exists. Without smoothing the average precision was between 0.43 and 0.49 across sizes; the frozen 3×3 mean lowered it to 0.03 at $b = 1$ and 0.31 at $b = 6$ (Table 4). Smoothing hurt at every size, the loss shrinking as the anomalies grew, and a 5×5 mean was worse than the 3×3 mean at every size (paired 95% intervals of all these differences exclude zero). The champion of the GaaR search was better than the frozen configuration for $b \leq 3$ and not distinguishable from it for $b \geq 4$, but it stayed below no smoothing at every size: configuration selection on the development scenarios, in which smoothing helped the DINOv2 images, mitigated the loss without removing it.

6. Limitations and future work

The lattices and the images are simulated and were written by the author of the method (the simulation is MyUncle's, with an easy kind of anomaly, and its parameter sets are not fitted to

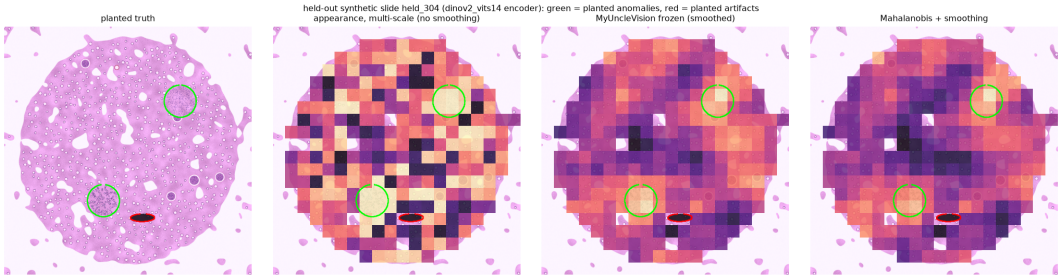


Figure 3: One held-out synthetic image (DINOv2): planted anomalies (green), planted artifact (red), the unsmoothed multi-scale appearance score, the frozen score and the smoothed Mahalanobis score on the patch lattice.

Table 3: Fresh test scenarios: average precision of the frozen configuration, of no smoothing and of the champions found on the development scenarios (development fitness F in column 2; test AP for handcrafted-feature images, DINOv2 images, and lattices with the rotor and the rotor + D embedding in columns 3–6; test fitness with a 95% interval in column 7, and its paired difference from the default in column 8; intervals resample the 16 images and the 16 simulation runs of each scenario).

Configuration	F_{dev}	Img. hand.	Img. DINOv2	Lat. rotor	Lat. rotor+D	F_{test} [95% interval]	Difference from default [95% interval]
Frozen default	0.26	0.04	0.50	0.32	0.46	0.18 [0.15, 0.39]	-
No smoothing	0.28	0.18	0.11	0.56	0.43	0.21 [0.15, 0.31]	+0.030 [−0.16, +0.08]
GaaR restart / population champion	0.35	0.17	0.41	0.41	0.45	0.26 [0.17, 0.42]	+0.080 [−0.05, +0.17]
GaaR greedy / random-search champion	0.33	0.24	0.12	0.41	0.44	0.21 [0.14, 0.33]	+0.030 [−0.17, +0.10]
Exhaustive development optimum	0.35	0.11	0.58	0.41	0.43	0.25 [0.18, 0.45]	+0.065 [−0.03, +0.19]

Table 4: Average precision against anomaly block size b (independent embedding-lattice generator; 3% anomalous sites, fixed separability).

Configuration	$b = 1$	2	3	4	6
No smoothing	0.44	0.43	0.45	0.49	0.43
Frozen (3×3 mean)	0.03	0.06	0.11	0.21	0.31
5×5 mean	0.03	0.04	0.04	0.07	0.16
GaaR restart champion	0.05	0.14	0.21	0.26	0.29
Exhaustive development optimum	0.04	0.09	0.13	0.20	0.25

measurements); nothing here shows that the method finds anything in an experimental system. The held-out sets were used once but are of the same family as the development sets, four runs per parameter set are few, the two embeddings disagree on whether smoothing helps, and a single Gaussian model scored higher than the frozen score on the development data. The configuration search is supervised, each strategy contributed one champion to the test comparison, and the test scenarios share their generators with the development data; the independent generator of Section 5.4 is still synthetic, and real images or a public data set are needed to judge the gain. Novelty is not a defect: a rare state may be a rare preparation, an instrument or a known feature that is merely uncommon. Planned: validation on experimental micrographs of ordered materials with independently established defects and on public histology, with settings, baselines and endpoints frozen and published before an independent set is analyzed once, blinded expert review, bootstrap intervals over images, a second synthetic generator, other spatial filters, and fitting the MyUncle parameter sets to measured data. If no advantage over the baselines appears, that result will be reported.

7. Conclusions

The local disorder and von Mises–Fisher order information of MyUncle generalize to lattices of unit vectors in d dimensions and equal MyUncle's at $d = 2$. A label-free score built on them was tested on simulated lattices and on synthetic images with frozen settings and four baselines. The results are mixed: smoothing helped only with the DINOv2 embeddings of the synthetic images, whose unsmoothed score is noisy, and hurt on simulated lattices and in a controlled study on an independent generator, least for compact anomalies; adding the local disorder as a feature helped on the simulated lattices, and a simple Mahalanobis baseline was a strong competitor. A supervised configuration search with GaaR improved the lower-tail average precision on fresh test scenarios, but the paired interval of the gain included zero and plain random search also beat the default, so any gain belongs to the selected configuration. All evidence is simulated; experimental validation is the next step.

Declarations

Funding: no external funding. **Competing interests:** the author declares no competing interests. **CRediT:** Leon Sandler, sole author: conceptualization, methodology, software, validation, formal analysis, investigation, visualization, writing – original draft, writing – review and editing. **Data and code availability:** code, tests, benchmarks and results are open at `https://github.com/sandlerleon/MyUncleVision` (MIT license) and archived at `https://doi.org/10.5281/zenodo.24066260` (all versions); MyUncle is vendored unmodified (`https://doi.org/10.5281/zenodo.21223570`). No experimental, patient or tissue data were used. **Generative AI:** during the preparation of this work the author used Claude (Anthropic) to

assist with writing and testing the code, running the analyses and drafting the text and figures; the author reviewed and edited the content and takes full responsibility for it.

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